

Large-gap non-Dirac quantum anomalous Hall effect in nitrided TlSb monolayers on nonmagnetic substrates

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Two-dimensional atomic monolayers of heavy elements grown on insulating substrates are promising for creating devices that exploit topological quantum states. Here, based on first-principles calculations and $k \cdot p$ model, we systematically investigate the electronic structures and topological properties of the TlSb monolayer epitaxially grown on nonmagnetic BaTe(111) surface, TlSb/BaTe(111). We discover a non-Dirac quantum spin Hall gap of 331.0 meV at its Γ point, induced by strong spin-orbit coupling (SOC) of the Sb and Tl p_x/p_y orbitals. Remarkably, nitrogen functionalization introduces ferromagnetic magnetization that breaks time-reversal symmetry and results in fully spin-polarized states, where the spin-up bands precisely degenerate at the Fermi energy with quadratic non-Dirac band dispersions. This spin-polarized non-Dirac bands, combined with SOC, opens a nontrivial gap of 258.0 meV with a significant Rashba effect, enabling the realization of a large-gap non-Dirac quantum anomalous Hall (QAH) effect, which is verified by a calculated Chern number of 1 and the observed topological edge states. We elucidate the underlying mechanism of this QAH state using the $k \cdot p$ model. Furthermore, our findings suggest that this approach can be extended to other nonmagnetic substrates such as SrTe and BaSe, demonstrating a versatile strategy to achieve large-gap QAH effects without the need for magnetic doping or specialized substrates.

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I. INTRODUCTION

The quantum anomalous Hall (QAH) effect, first theoretically proposed by Haldane [1], is a novel two-dimensional (2D) topological quantum state. The QAH state is insulating in the bulk but exhibits robust conducting edge states that are topologically protected from backscattering, leading to quantized Hall conductivity without the need of an external magnetic field [1–4]. Realizing QAH effect in realistic material systems has attracted significant attention in condensed matter physics and materials science due to its immense potential for applications in low-power-consumption spintronics and electronic devices [3,4]. To date, several strategies based on different band structure types, such as Dirac bands and non-Dirac bands, have been proposed to achieve the QAH effect [5–14]. For nonmagnetic Dirac bands, the Kane-Mele model [15] shows that the atomic spin-orbit coupling (SOC) can open a topologically nontrivial gap at the degenerate Dirac points. With the introduction of a magnetic exchange field, the Dirac bands become spin polarized, facilitating the intersection of spin-up and spin-down bands near the Dirac point, where the QAH gap can subsequently be established via Rashba SOC [5–8]. When the Dirac bands are fully spin-polarized, resulting in complete separation of spin-up and spin-down bands, a finite atomic SOC can induce a QAH

gap at the single-spin Dirac point with a Chern number of ± 1 [9]. Unlike Dirac bands, non-Dirac bands with full spin-polarizations typically exhibit a trivial band gap, necessitating a significant SOC to inverse such a gap and consequently achieve the QAH effect [10,16]. To circumvent the need for large SOC, it is advantageous to have gapless non-Dirac bands characterized by quadratic band degeneracy, which serve as critical points for topological transitions. In this context, a finite atomic SOC can induce a topological gap around the transition point, resulting in a QAH effect with Chern number of ± 1 [11–14]. This non-Dirac QAH state is robust due to the protection provided by crystal symmetry and exhibits a relatively large nontrivial gap. Furthermore, gapless non-Dirac band materials require lower crystal symmetry compared to Dirac band materials [12], making it easier to realize gapless non-Dirac band structures in realistic material systems. Therefore, designing large-gap QAH state based on gapless non-Dirac bands is of great significance.

Based on the strategies described above, researchers have theoretically predicted some QAH materials [17–34], such as transition-metal (TM) atom-doped topological insulator thin films [17], graphene- and silicene-based systems [18–21], monolayer TM trihalides MX_3 [22–28], heterostructures of heavy atomic layers on magnetic insulators [29], monolayer TM oxides [30], layered MnBi_2Te_4 -family materials [31,32], and so on. Notably, experimental observations of the QAH effect have been achieved in several material systems, including TM atom-doped $(\text{Bi}, \text{Sb})_2\text{Te}_3$ thin films [35,36], layered MnBi_2Te_4 [37], and graphene and transition-metal

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dichalcogenide (TMD)-based moiré materials [38–43], significantly advancing the development of this field. Although the QAH state has been realized experimentally, the nontrivial gap remains too small for practical applications. Hence, the search and design of QAH materials with large topologically nontrivial gap is of great significance. Heavy atoms exhibit strong atomic SOC, which is advantageous for designing topological quantum states with large nontrivial gaps. Functionalized 2D honeycomb atomic monolayers composed of heavy elements including those of groups IV [44–46], V [47–52], and III-V [53,54] have been predicted to harbor the large-gap quantum spin Hall (QSH) and QAH states. However, achieving these functionalized 2D atomic monolayers with topologically nontrivial states in a freestanding form is quite challenging in practice. Therefore, epitaxial growth of 2D atomic monolayers on an insulating substrate presents a viable solution for stabilizing these materials. For the QSH state, several theoretical studies [55–58] predicted that it could be realized in functionalized 2D atomic monolayers grown on nonmagnetic insulating substrate systems. While for the QAH state, although heterostructures of 2D atomic monolayers on magnetic insulating substrate can be used to generate the QAH effect, the magnetic proximity effect may be relatively weak, and magnetic substrates might disrupt the topological band structures. If the QAH states can be realized in nonmagnetic substrate systems, the potential adverse effects caused by magnetic substrates may be eliminated. Hence, designing the large-gap QAH state in the nonmagnetic substrate-supported atomic monolayer systems is of great significance and is experimentally feasible.

In this work, we systematically investigate the electronic structures and topological properties of the heterostructure material system composed of the TlSb monolayer epitaxially grown on the nonmagnetic BaTe(111) surface with first-principles calculations and $k \cdot p$ model methods. By performing band structure calculations without including SOC, we show that the degenerate energy point with quadratic non-Dirac band dispersion emerges at the Γ point in TlSb/BaTe(111), mainly contributed by p_x/p_y orbitals of the Sb and Tl atoms. A large-gap (331.0 meV) QSH state is obtained around the non-Dirac Γ point due to the elimination of degeneracy by including SOC. However, TlSb/BaTe(111) is in a metallic state because the Fermi energy (E_F) is not located within the obtained QSH gap. Excitingly, our calculations demonstrate that the magnetism can be achieved in TlSb/BaTe(111) through nitrogen functionalization, and the magnetic ground state of the N-TlSb/BaTe(111) is in the ferromagnetic (FM) state. Notably, the non-Dirac bands become spin polarized in N-TlSb/BaTe(111) before SOC is turned on, with the spin-up non-Dirac bands being precisely degenerate at the E_F and the Γ point, after SOC is included, a large topologically nontrivial band gap of 258.0 meV is opened around the spin-up non-Dirac Γ point, with the E_F located exactly within the nontrivial gap. The calculated Chern number, anomalous Hall conductivity, and edge states confirm the obtained intrinsic large-gap non-Dirac QAH state in the N-TlSb/BaTe(111) system. Moreover, a significant Rashba SOC, induced by inversion asymmetry, is observed in N-TlSb/BaTe(111). The topological mechanism and band evolution of the obtained non-Dirac topologically nontrivial

state with significant Rashba SOC is analyzed through a $k \cdot p$ model. Furthermore, our calculations indicate that large-gap non-Dirac QAH states also exist in the FM states of the N-TlSb/SrTe(111) and N-TlSb/BaSe(111) systems.

II. COMPUTATIONAL DETAILS

The geometry and electronic structures of the studied heterostructures (N)-TlSb/BaTe(111) are calculated by using the projected augmented wave (PAW) [59] formalism based on density functional theory (DFT), as implemented in the Vienna *Ab initio* Simulation Package (VASP) [60]. The Perdew-Burke-Ernzerhof generalized gradient approximation (GGA-PBE) is employed to describe the exchange and correlation functional [61]. In addition, the hybrid functional HSE06 [62] method, known for its higher accuracy in calculation results, is also employed to calculate the electronic band structures and topological properties of the studied heterostructures. The plane-wave cutoff energy is set to be 550 eV, and the convergence criteria for the total energy is 10^{-6} eV. A vacuum space of larger than 20 Å is used to avoid any interaction between adjacent slabs. An $18 \times 18 \times 1$ Γ central Monkhorst-Pack grids is used for the unit cell calculations. There are nine atomic layers of substrate in the constructed heterostructure systems. To stabilize the substrate and simulate bulk surface, the one Te terminal surface of the BaTe(111) film, opposite to the TlSb monolayer interface, is passivated by hydrogen atoms. In addition, the bottom five atomic layers in BaTe(111) film are fixed at the bulk crystal structure. Except for the five bottom atomic layers of the BaTe(111) substrate, which are kept fixed during structural relaxation, all other atoms in the constructed heterostructures are allowed to relax until the Hellmann-Feynman force on each atom is smaller than 0.01 eV/Å. To investigate the topological properties of the constructed heterostructures, the maximally localized Wannier functions (MLWFs) [63,64] are constructed by using the WANNIER90 package [65]. The Berry curvatures, anomalous Hall conductivity, and Chern numbers are calculated by employing the algorithm described in Ref. [66]. The Z_2 topological invariants and edge states are calculated by using the WANNIERTOOLS package [67].

III. RESULTS AND DISCUSSION

A. Non-Dirac QSH state in TlSb/BaTe(111)

The 2D TlSb atomic monolayer is in the buckled honeycomb lattice with an optimized lattice constant of 4.82 Å, as shown in Fig. S1(a) in the Supplemental Material (SM) [68]. The pristine freestanding TlSb monolayer has a small band gap around the Γ point when excluding the SOC, and a nontrivial band gap (~ 0.3 eV) opens due to the SOC-induced band inversion, as illustrate in Fig. S1 in the SM [68]. Although TlSb monolayer is a large-gap QSH insulator [69], experimental synthesis of this kind of 2D atomic monolayer in a freestanding form is full of challenging. An experimentally feasible scheme for stabilizing such atomic monolayer is epitaxial growth of it on an insulating substrate surface with hexagonal lattice, such as the experimental work of bimuthene/SiC [70]. The bulk BaTe has an insulating band gap, and the BaTe (111) surface is in hexagonal lattice with

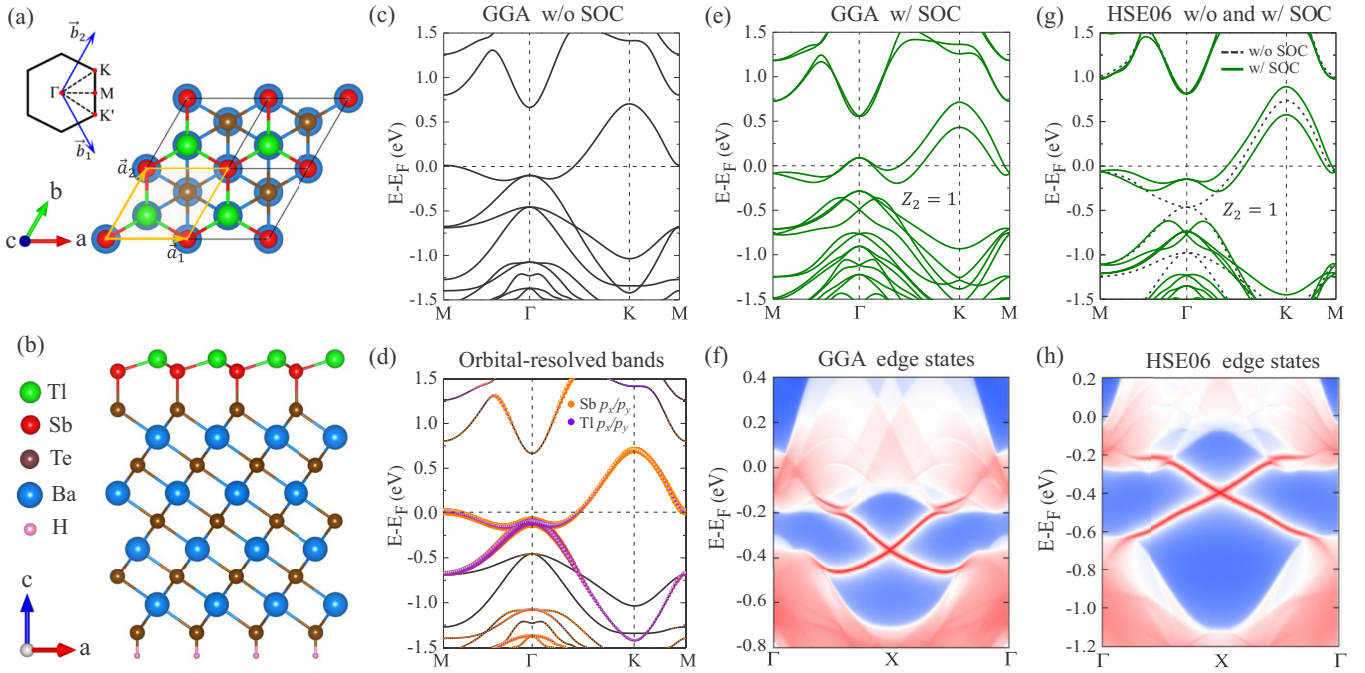


FIG. 1. (a) Top view and (b) side view of the Te-terminated BaTe(111)-supported TlSb monolayer, denoted as TlSb/BaTe(111). The rhombus in (a) shows the unit cell with lattice vectors $\vec{a}_1$ and $\vec{a}_2$. The inset in (a) shows the first Brillouin zone with high symmetry points. (c) Band structure of TlSb/BaTe(111) without SOC obtained by using GGA-PBE calculations. (d) Orbital-resolved band structure of (c), where the contributions from Sb and Tl p_x/p_y states to the bands are denoted by orange and violet dots, respectively. (e) The corresponding band structure of (c) with the consideration of SOC. (f) Density of states of semi-infinite TlSb/BaTe(111) obtained by using GGA-PBE calculations. (g) Band structures of TlSb/BaTe(111) without SOC (black dashed line) and with SOC (olive solid line) obtained by using HSE06 calculations. (h) The same as (f), except that the density of states are calculated by HSE06 method.

experimental lattice constant of 4.95 Å, which is close to the in-plane lattice constant of TlSb monolayer. Therefore, the nonmagnetic BaTe(111) surface is an appropriate insulating substrate that can support the TlSb monolayer. To investigate the stable structure of the heterostructure of TlSb monolayer grown on the Te-terminated BaTe(111) surface, we consider the following four typical configurations: Type-A, Type-B, Type-C, and Type-D, as shown in Fig. S2 in the SM [68]. For the Type-A (Type-B) and Type-C (Type-D) configurations, the TlSb/BaTe(111) interface consists of having the Sb and Tl atoms on top of the Te atoms, respectively. The binding energy E_b of the built TlSb/BaTe(111) heterostructure is calculated as $E_b = E_{\text{TlSb}} + E_{\text{BaTe(111)}} - E_{\text{Total}}$, where E_{TlSb} , $E_{\text{BaTe(111)}}$, and E_{Total} are the total energies of the TlSb monolayer, BaTe(111) substrate, and TlSb/BaTe(111) heterostructure, respectively. The obtained binding energies for the TlSb/BaTe(111) heterostructures with Type-A, Type-B, Type-C, and Type-D configurations are 1.615 eV, 1.595 eV, 1.191 eV, 1.170 eV, respectively, from which we can observe that the Type-A and Type-B have almost the same binding energy and are the stable combination configurations. Therefore, the TlSb/BaTe(111) heterostructure with Type-A and Type-B configurations are employed in the following investigations. The calculation results for Type-B configuration are given in the SM [68].

In Figs. 1(a) and 1(b) we show the top view and side view of the optimized atomic structure of TlSb/BaTe(111) with Type-A configuration employed in our calculations.

In the following sections, except for special indication, TlSb/BaTe(111) refers to TlSb/BaTe(111) with a Type-A configuration. In the optimized TlSb/BaTe(111) structure, the obtained Tl-Sb and Sb-Te bond lengths are 2.99 Å and 2.76 Å, respectively, the buckling distance between Tl and Sb is $\Delta = 0.90$ Å. Figure 1(c) shows the band structure without SOC for TlSb/BaTe(111) obtained by using GGA-PBE calculations, from which we can observe that the two bands near the E_F are degenerate at the Γ point with quadratic non-Dirac band dispersions. Figure 1(d) presents the orbital-resolved band structure of Fig. 1(c), illustrating that the two non-Dirac bands near the Γ point and E_F are mainly contributed by the p_x/p_y states of Sb and Tl atoms. The emergence of non-Dirac energy degeneracy at the Γ point is due to the static configuration of TlSb/BaTe(111) possessing threefold rotation symmetry (C_3) along the c axis. Therefore, in the TlSb/BaTe(111) heterostructure, the two non-Dirac bands originating from the TlSb monolayer are located within the insulating band gap of the BaTe(111) substrate, as shown in Fig. 1(c). Figure 1(e) displays the corresponding band structure of Fig. 1(c) with SOC turned on, showing that an indirect band gap (87.7 meV) is opened around the non-Dirac degenerate Γ point by introducing SOC.

To identify the topological properties of the SOC-induced band gap in Fig. 1(e), the Z_2 topological invariant and edge states of the TlSb/BaTe(111) are calculated. When the E_F is tuned into the SOC-induced band gap in Fig. 1(e), the obtained Z_2 topological invariant is $Z_2 = 1$, indicating this

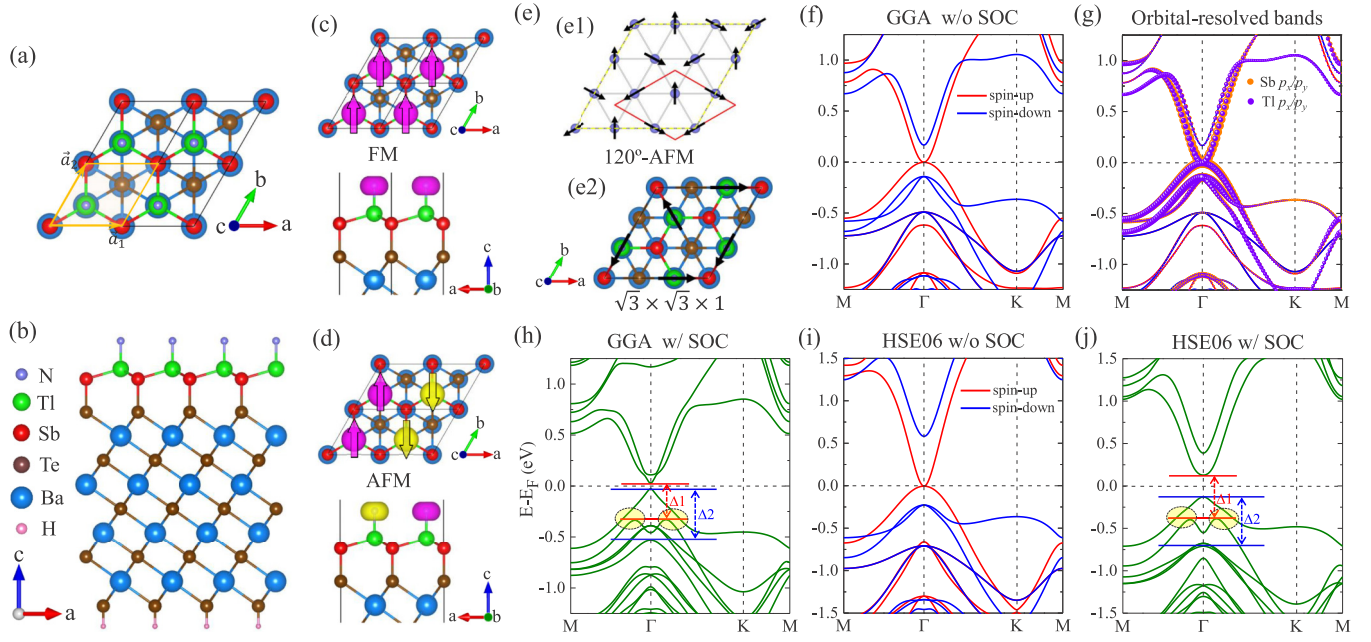


FIG. 2. (a) Top view and (b) side view of the nitrogen functionalized TlSb/BaTe(111), denoted as N-TlSb/BaTe(111). (c), (d) Spatial spin density distributions ($\rho_{\uparrow} - \rho_{\downarrow}$): top and side views of the FM (c) and collinear AFM (d) magnetic configurations. The magenta and yellow arrows in (c) and (d) denote the up and down spins, respectively. (e) The noncollinear 120°-AFM magnetic configuration. The yellow and red rhombus in (e1) denote the $3 \times 3 \times 1$ and $\sqrt{3} \times \sqrt{3} \times 1$ supercell of the 2D triangular lattice. (e2) The $\sqrt{3} \times \sqrt{3} \times 1$ supercell of N-TlSb/BaTe(111) with 120°-AFM. The black arrows in (e) denote the noncollinear spins that lie in the 2D plane. (f) Spin-polarized band structure of N-TlSb/BaTe(111) obtained by using GGA-PBE calculations. The red and blue curves denote the spin-up and spin-down bands, respectively. (g) Orbital-resolved band structures of (f), where the contributions from Sb and Tl p_x/p_y states to the bands are denoted by orange and violet dots, respectively. (h) The corresponding band structure of (c) with the consideration of SOC. (i), (j) Band structures of N-TlSb/BaTe(111) without SOC (i) and with SOC (j) obtained by using HSE06 calculations.

insulating state is the QSH state. Figure 1(f) plots the calculated density of states (DOS) of the semi-infinite TlSb/BaTe(111) system, showing that two topologically nontrivial edge states exist inside the SOC-induced gap, confirming the obtained QSH gap with $Z_2 = 1$. Figure 1(g) shows the band structures of TlSb/BaTe(111) without SOC (black dashed line) and with SOC (olive solid line) obtained by using HSE06 calculations, illustrating that the two non-Dirac bands still degenerate at the Γ point when SOC is not considered, and an indirect band gap of 331.0 meV is opened when SOC is included. The calculated Z_2 topological invariant of $Z_2 = 1$ and the topologically nontrivial edge states shown in Fig. 1(h) confirm the QSH gap obtained from the HSE06 calculations. Although the large-gap QSH state is obtained, the TlSb/BaTe(111) heterostructure is in a metallic state due to the E_F is not located within the QSH gap. The calculated electronic band structures and topological properties of the TlSb/BaTe(111) with Type-B configuration are given in Fig. S4 in the SM [68], demonstrating that the large-gap QSH state still exists in Type-B configuration. To achieve the QAH state in the designed TlSb/BaTe(111) heterostructure, the two non-Dirac bands need to be tuned to become spin polarized, with only one spin state existing near the E_F , by introducing magnetism into this system. In the following section, the nitrogen atoms are used to tune the magnetism, electronic structures, and topological properties of the TlSb/BaTe(111), aiming to realize a large-gap QAH state.

B. Intrinsic large-gap non-Dirac QAH state in N-TlSb/BaTe(111)

To remove the unsaturated dangling bond states in TlSb/BaTe(111), we passivate the exposed surface Tl atoms with nitrogen atoms. Figures 2(a) and 2(b) show the top view and side view of the nitrogen functionalized TlSb/BaTe(111) [denoted as N-TlSb/BaTe(111)] with optimized atomic structure. Our calculations illustrate that the nitrogen atoms form strong bonds with the surface Tl atoms, with a large binding energy of 1.09 eV per unit cell. Remarkably, we find that the magnetism can be introduced in the TlSb/BaTe(111) heterostructure through nitrogen functionalization. Our first-principles calculations show that the FM state is more stable than the nonmagnetic (NM) state by 1306.45 meV per unit cell, indicating that the spin-polarized state is more stable for N-TlSb/BaTe(111). For the FM state, the net magnetic moment per unit cell is found to be $2.0 \mu_B$, and the resultant localized electrons on the N atoms contribute to this net magnetic moment, as clearly illustrated by the calculated spin density distribution shown in Fig. 2(c). To investigate the magnetic ground state of N-TlSb/BaTe(111), we consider the following three typical magnetic configurations: FM, collinear antiferromagnetic (AFM), and noncollinear 120°-AFM, as depicted in Figs. 2(c)–2(e). Our first-principles calculations show that the magnetic ground state of the N-TlSb/BaTe(111) system is in the FM state, which is more stable than the collinear AFM state and noncollinear 120°-AFM state by 2.65 meV and 2.92 meV per unit cell, respectively. The

magnetic anisotropy energy (MAE) per N atom for the N-TISb/BaTe(111) system is defined as $\text{MAE} = E_{[100]} - E_{[001]}$, where $E_{[100]}$ and $E_{[001]}$ are the total energies of the FM state N-TISb/BaTe(111) unit cell with the magnetization along the [100] and [001] directions, respectively. The calculated MAE for N-TISb/BaTe(111) is 1.03 meV per N atom. The obtained positive value of MAE ($\text{MAE} > 0$) indicates that the easy axis of magnetization for the N-TISb/BaTe(111) system is oriented along the c -axis direction (out-of-plane). In addition, the stability of the FM state can be enhanced by applying a small vertical external magnetic field.

Figure 2(f) displays the spin-polarized band structure of N-TISb/BaTe(111) calculated using the GGA-PBE method, from which we can observe that the non-Dirac bands around E_F are spin polarized due to the introduction of FM order. Close to E_F in Fig. 2(f), we can find one spin-up and one spin-down non-Dirac bands touching points at the Γ point. The orbital-resolved band structures provided in Fig. 2(g) reveal that the two spin-up and two spin-down non-Dirac bands near E_F are dominated by the p_x/p_y orbitals of the Sb and Tl atoms. In the spin-up state, the two parabolic non-Dirac bands exhibit opposite opening directions, whereas in the spin-down state, both parabolic non-Dirac bands open downward, resulting in a spin-gapless half-metallic state with only the spin-up non-Dirac bands being precisely degenerate at both the E_F and the Γ point. When SOC is included, the energy gaps will be opened at both the spin-up and spin-down non-Dirac bands degenerate Γ point. Figure 2(h) plots the corresponding band structure of Fig. 2(f) with the consideration of SOC. It illustrates that the SOC-induced indirect band gap around the Γ point for the two spin-up non-Dirac bands is $\Delta_1 = 354.5$ meV, and the SOC-induced energy gap at the Γ point for the two spin-down non-Dirac bands is $\Delta_2 = 505.5$ meV. Because one spin-down non-Dirac band shifts into the SOC-induced gap Δ_1 of the spin-up non-Dirac bands, the global band gap in Fig. 2(h) is 50.0 meV, with the E_F located exactly within the global band gap. In addition, a significant Rashba SOC is observed in N-TISb/BaTe(111), as highlighted in the yellow shaded areas in Fig. 2(h). Figure 2(i) shows the spin-polarized band structure of N-TISb/BaTe(111) calculated by using the HSE06 functional, indicating that the obtained two featured spin-up and spin-down non-Dirac bands still exist. The HSE06 band structure of N-TISb/BaTe(111) with SOC included is plotted in Fig. 2(j), showing that the SOC-induced band gap for the two spin-up non-Dirac bands and energy gap for the two spin-down non-Dirac bands are $\Delta_1 = 513.5$ meV and $\Delta_2 = 580.0$ meV, respectively. Furthermore, the SOC-included global band gap in N-TISb/BaTe(111) calculated by using HSE06 method is 258.0 meV, this global band gap is much larger than that obtained by using GGA-PBE method due to the downward shift of the spin-down non-Dirac bands, as shown in Fig. 2(j).

The Berry curvatures $[\Omega(\mathbf{k})]$ and Chern numbers are calculated to identify the topological properties of the SOC-induced global band gap in the constructed N-TISb/BaTe(111) system. The Berry curvatures are calculated by [71,72]

$$\Omega(\mathbf{k}) = \sum_n f_n \Omega_n(\mathbf{k}), \quad (1)$$

$$\Omega_n(\mathbf{k}) = -2\text{Im} \sum_{m \neq n} \frac{\hbar^2 \langle \psi_{n\mathbf{k}} | v_x | \psi_{m\mathbf{k}} \rangle \langle \psi_{m\mathbf{k}} | v_y | \psi_{n\mathbf{k}} \rangle}{(E_m - E_n)^2}, \quad (2)$$

where f_n is the Fermi-Dirac distribution function, $v_{x(y)}$ are the velocity operators, E_n is the eigenvalue of the Bloch functions $|\psi_{n\mathbf{k}}\rangle$, and the summation is over all the occupied states. The Chern number C can be obtained by integrating the Berry curvatures over the first Brillouin zone, as $C = \frac{1}{2\pi} \sum_n \int_{\text{BZ}} d^2k \Omega_n$. Figure 3(a) shows the magnified GGA-PBE band structures of Fig. 2(f) (spin-polarized bands) and Fig. 2(h) (SOC bands) around the E_F . The violet dots in Fig. 3(a) display the calculated $\Omega(\mathbf{k})$ of N-TISb/BaTe(111) with the E_F located inside the SOC-induced global non-Dirac band gap. We can observe that the large values of the obtained $\Omega(\mathbf{k})$ mainly distribute around the non-Dirac Γ point. By integrating the calculated $\Omega(\mathbf{k})$ over the first BZ, the nonzero integer Chern number of $C = 1$ is obtained, indicating the SOC-induced global non-Dirac band gap in Fig. 2(h) is the QAH state. Figure 3(b) plots the calculated anomalous Hall conductivity (σ_{xy}) as a function of the E_F position, showing that a quantized anomalous Hall conductivity platform with $C = 1$ emerges around the original E_F , demonstrating that N-TISb/BaTe(111), with GGA-PBE calculations, is indeed an intrinsic QAH insulator, with E_F located exactly inside the global non-Dirac nontrivial gap. The density of states (DOS) of the semi-infinite N-TISb/BaTe(111) system is plotted in Fig. 3(c), showing one topologically nontrivial edge state that connects the valence and conduction band, confirming the achieved intrinsic QAH state with $C = 1$. The calculated Berry curvatures, anomalous Hall conductivity, Chern number, and topologically nontrivial edge states presented in Figs. 3(d)–3(f) illustrate that the SOC-induced global non-Dirac band gap in Fig. 2(j) for N-TISb/BaTe(111) with HSE06 calculations is an intrinsic large-gap (258.0 meV) non-Dirac QAH state with $C = 1$. The calculated electronic band structures, Berry curvatures, anomalous Hall conductivity, and Chern number of the N-TISb/BaTe(111) with Type-B configuration are displayed in Fig. S5 in the SM [68], illustrating that the intrinsic non-Dirac QAH state with significant Rashba SOC still exists in Type-B configuration. Therefore, both the GGA-PBE and HSE06 calculations demonstrate the realization of an intrinsic large-gap non-Dirac QAH state in the constructed N-TISb/BaTe(111) system.

To better understand the band evolution and topological mechanism of the spin-polarized non-Dirac bands in N-TISb/BaTe(111), the $k \cdot p$ model Hamiltonian is employed. Since the states around the spin-up and spin-down non-Dirac energy degenerate Γ point are both dominated by p_x and p_y orbitals, as illustrated in Fig. 2(g), it is reasonable to adopt p_x and p_y orbitals as the basis of the $k \cdot p$ model Hamiltonian. For convenience, the two orbitals can be transformed to the basis of $\{|p_+, \uparrow\rangle, |p_-, \uparrow\rangle, |p_+, \downarrow\rangle, |p_-, \downarrow\rangle\}$ with $|p_{\pm}\rangle = \mp \frac{1}{\sqrt{2}}(|p_x \pm i p_y\rangle)$ and $\uparrow(\downarrow)$ for the spin-up (spin-down) state. With the consideration of the C_3 symmetry of the p_x and p_y orbitals, the total effective $k \cdot p$ model Hamiltonian including the hopping term, atomic SOC term, Rashba SOC term, and exchange field term can be written

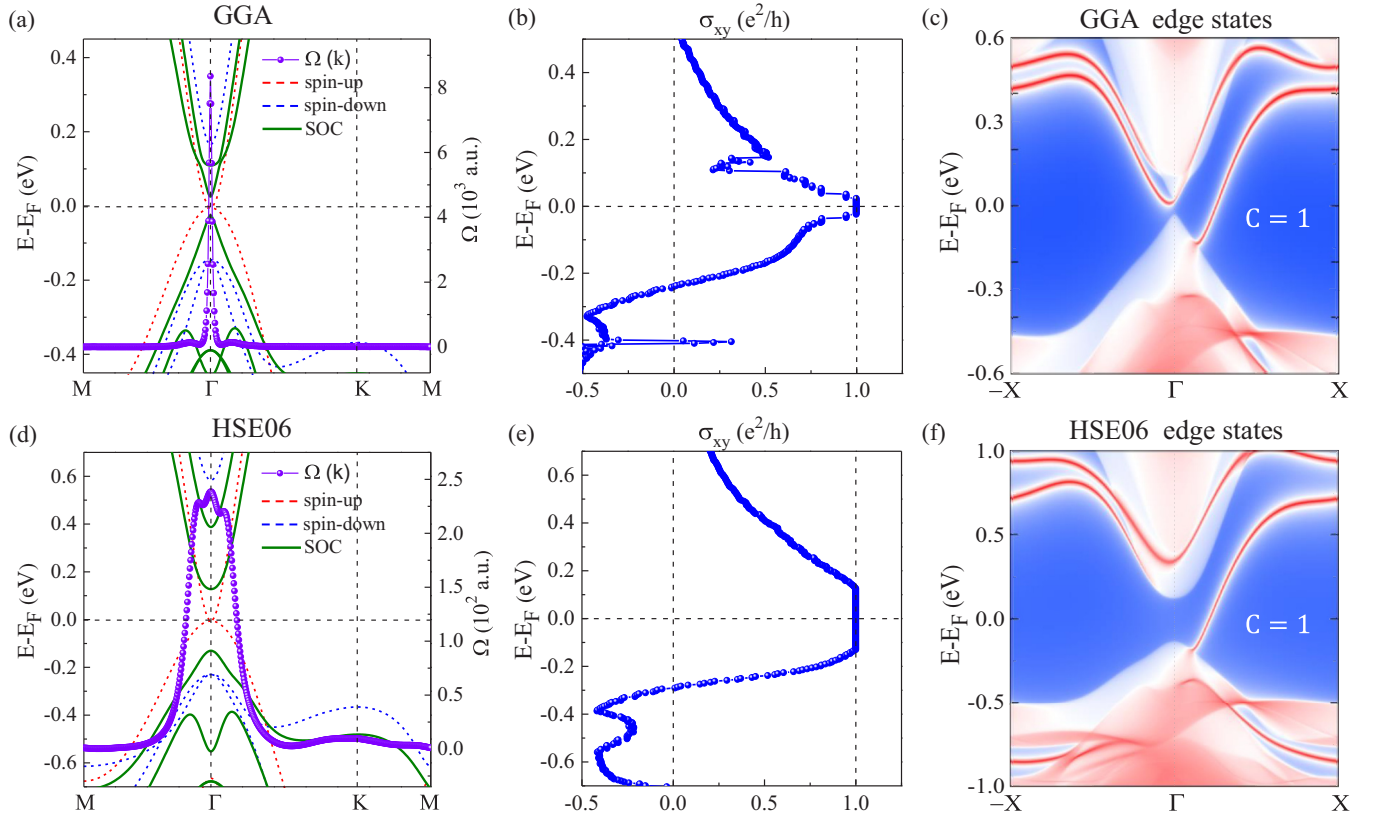


FIG. 3. (a) The magnified spin-polarized (red and blue dashed curves) and SOC (olive curves) band structures of the N-TISb/BaTe(111) around the E_F obtained from GGA-PBE calculations. The red and blue dashed curves denote the spin-up and spin-down bands, respectively. The violet dots in (a) denote the calculated Berry curvatures. (b) The anomalous Hall conductivity σ_{xy} as a function of the position of E_F calculated by using GGA-PBE method. (c) Edge states of semi-infinite N-TISb/BaTe(111) obtained by using GGA-PBE calculations. (d)–(f) The same as (a)–(c), except that the band structures, anomalous Hall conductivity, and edge states are calculated by using HSE06 method.

as [11]

$$\begin{aligned}
 H(k) &= H_0(k) + H_{\text{soc}}(k) + H_R(k) - H_M(k) \\
 &= \begin{bmatrix} \alpha_1 k^2 & \beta_1 k_-^2 & 0 & 0 \\ \beta_1^* k_+^2 & \alpha_1 k^2 & 0 & 0 \\ 0 & 0 & \alpha_2 k^2 & \beta_2 k_-^2 \\ 0 & 0 & \beta_2^* k_+^2 & \alpha_2 k^2 \end{bmatrix} \\
 &+ \begin{bmatrix} \lambda_{\text{soc}} & 0 & 0 & 0 \\ 0 & -\lambda_{\text{soc}} & 0 & 0 \\ 0 & 0 & -\lambda_{\text{soc}} & 0 \\ 0 & 0 & 0 & \lambda_{\text{soc}} \end{bmatrix} \\
 &+ \begin{bmatrix} 0 & 0 & i\lambda_R k_- & 0 \\ 0 & 0 & i\lambda_R' k_+ & i\lambda_R k_- \\ -i\lambda_R^* k_+ & -i\lambda_R^* k_- & 0 & 0 \\ 0 & -i\lambda_R^* k_+ & 0 & 0 \end{bmatrix} \\
 &- \begin{bmatrix} M & 0 & 0 & 0 \\ 0 & M & 0 & 0 \\ 0 & 0 & -M & 0 \\ 0 & 0 & 0 & -M \end{bmatrix}, \quad (3)
 \end{aligned}$$

where H_0 expresses the hopping term, α_1 and β_1 are the parameters for the spin-up subspace, α_2 and β_2 are the parameters for the spin-down subspace, H_{soc} is the atomic SOC term and λ_{soc} is the atomic SOC strength, H_R is the Rashba SOC term and λ_R and λ_R' are the strength of Rashba SOC, H_M is the exchange field term, and the M is the exchange field strength.

Figure 4(a) plots the calculated spin-polarized $k \cdot p$ model bands with parameters of $\alpha_1 = 1.50$ eV, $\beta_1 = 3.00$ eV, $\alpha_2 = -3.00$ eV, $\beta_2 = 1.00$ eV, $M = -0.10$ eV, $\lambda_{\text{soc}} = 0.00$ eV, $\lambda_R = 0.00$ eV, and $\lambda_R' = 0.00$ eV. Due to the C_3 symmetry of the multiple p (p_x and p_y) orbitals, degenerate energy points with quadratic non-Dirac band touching appear at the Γ point for both spin-up and spin-down bands, as shown in Fig. 4(a). Additionally, the two spin-up parabolic non-Dirac bands exhibit opposite opening directions, while the two spin-down parabolic non-Dirac bands both open downward. This demonstrates that the band dispersions of the two spin-up and two spin-down non-Dirac bands obtained from $k \cdot p$ model calculations in Fig. 4(a) match the two spin-up and two spin-down non-Dirac bands around E_F obtained from DFT calculations in Figs. 2(f) and 2(i). Therefore, we can conclude that the spin-gapless half-metallic states in Figs. 2(f) and 2(i) obtained from DFT calculations can be well explained by the $k \cdot p$ model. When atomic SOC is further included, the indirect band gap Δ_1 around the Γ point is opened for the two spin-up

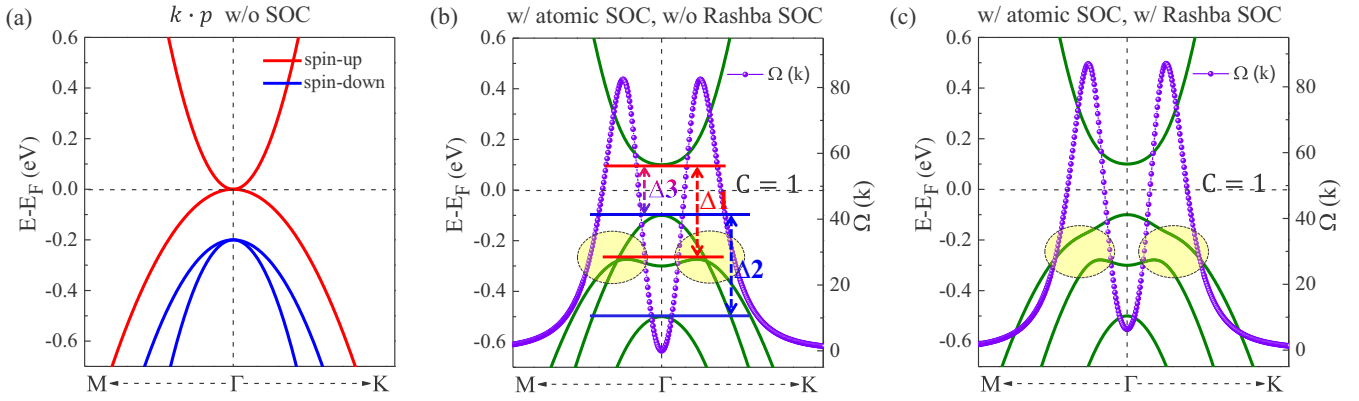


FIG. 4. (a) The calculated spin-polarized $k \cdot p$ model bands around the Γ point with the parameters: $\alpha_1 = 1.50$ eV, $\beta_1 = 3.00$ eV, $\alpha_2 = -3.00$ eV, $\beta_2 = 1.00$ eV, $M = -0.10$ eV, $\lambda_{\text{soc}} = 0.00$ eV, $\lambda_R = 0.00$ eV, and $\lambda'_R = 0.00$ eV. (b) The corresponding band structure of (a) with atomic SOC and without Rashba SOC by setting $\lambda_{\text{soc}} = 0.20$ eV, $\lambda_R = 0.00$ eV, and $\lambda'_R = 0.00$ eV. (c) The corresponding band structure of (a) with atomic SOC and with Rashba SOC by setting $\lambda_{\text{soc}} = 0.20$ eV, $\lambda_R = 0.40$ eV, and $\lambda'_R = -0.50$ eV. The violet dots in (b) and (c) denote the calculated Berry curvatures.

non-Dirac bands, and the energy gap Δ_2 at the Γ point is opened for the two spin-down non-Dirac bands, in addition, the atomic SOC-induced global band gap is Δ_3 , with the E_F located exactly within the global band gap, as illustrated in Fig. 4(b) with $\lambda_{\text{soc}} = 0.20$ eV. The violet dots in Fig. 4(b) plot the calculated $\Omega(\mathbf{k})$ when the E_F is located inside the atomic SOC-induced global band gap Δ_3 , in which the obtained $\Omega(\mathbf{k})$ around the Γ point are positive. Integrating the $\Omega(\mathbf{k})$ in the first BZ, the nonzero integer Chern number of $C = 1$ can be obtained, indicating the atomic SOC-induced global band gap Δ_3 in Fig. 4(b) is a QAH state, which is the same as the QAH states obtained from DFT calculations in Figs. 2(h) and 2(j).

The bands within the yellow shaded areas in Fig. 4(b) show band degeneracy points, resulting from the crossing of the spin-up and spin-down non-Dirac bands induced by atomic SOC. In contrast, the DFT bands in Figs. 2(h) and 2(j) do not exhibit any band degeneracy points within the corresponding yellow shaded areas for the spin-up and spin-down non-Dirac bands. Based on the $k \cdot p$ model bands shown in Fig. 4(b), when the Rashba SOC is further considered, the band degeneracy points in the yellow shaded areas in Fig. 4(b) will be eliminated, as illustrated in Fig. 4(c) with $\lambda_R = 0.40$ eV and $\lambda'_R = -0.50$ eV. By comparing Fig. 4(c) and Fig. 4(b), we can see that the introduction of Rashba SOC opens energy gaps at the crossing points of the spin-up and spin-down non-Dirac bands. The violet dots in Fig. 4(c) display the calculated $\Omega(\mathbf{k})$ around the Γ point. Integrating the calculated $\Omega(\mathbf{k})$ in the first BZ, the Chern number of $C = 1$ is obtained, illustrating the global band gap in Fig. 4(c) is also a QAH state, and its topologically nontrivial property does not destroyed by the presence of Rashba SOC. The characteristics of the DFT non-Dirac bands around E_F in Figs. 2(h) and 2(j) are the same as the $k \cdot p$ model bands in Fig. 4(c), indicating the existence of significant Rashba SOC in the N-TISb/BaTe(111) system. Therefore, the spin-polarized non-Dirac band structures and QAH state with significant Rashba SOC obtained in N-TISb/BaTe(111) can be well understood by the above constructed $k \cdot p$ model Hamiltonian.

C. Topological properties of N-TISb/SrTe(111) and N-TISb/BaSe(111)

Bulk SrTe and BaSe both have insulating band gaps and possess the same crystal structure as bulk BaTe. The experimental lattice constants of the SrTe(111) and BaSe(111) surface are 4.71 Å and 4.67 Å, respectively, which are also close to the in-plane lattice constant of TISb monolayer. Therefore, we further investigate the electronic structures, magnetic properties, and topological properties of the heterostructure formed by the epitaxial growth of the TISb monolayer on the Te-terminated SrTe(111) and Se-terminated BaSe(111) surface. Our calculations show that the Type-A and Type-B have almost the same binding energy and are the stable combination configurations for the TISb/SrTe(111) and TISb/BaSe(111) heterostructures. Notably, the magnetism can also be introduced in TISb/SrTe(111) and TISb/BaSe(111) through nitrogen functionalization. For N-TISb/SrTe(111) and N-TISb/BaSe(111), the FM states are more stable than the NM states by 1244.07 meV and 1237.87 meV per unit cell, respectively. This demonstrates that their spin-polarized states are more stable. Different from N-TISb/BaTe(111), the magnetic ground states of N-TISb/SrTe(111) and N-TISb/BaSe(111) are both collinear AFM states, which are more stable than the FM states by 1.18 meV and 2.80 meV, respectively. Due to the very small energy difference between the FM and AFM states, it is possible to tune the magnetic state of N-TISb/SrTe(111) and N-TISb/BaSe(111) into a stable FM state by applying a small vertical external magnetic field. Additionally, the Zeeman splitting energy is extremely small, on the order of 10^{-4} eV under a magnetic field of 1 T, indicating that applying a small external magnetic field will not affect the bands topological properties of N-TISb/SrTe(111) and N-TISb/BaSe(111) in the FM state.

Figures 5(a) and 5(d) show the spin-polarized band structures for the FM states of N-TISb/SrTe(111) and N-TISb/BaSe(111) with Type-A configuration, respectively, calculated using the HSE06 method. In Figs. 5(a) and 5(d), around the E_F , the two spin-up non-Dirac bands degenerate at the Γ point, indicating that before including SOC,

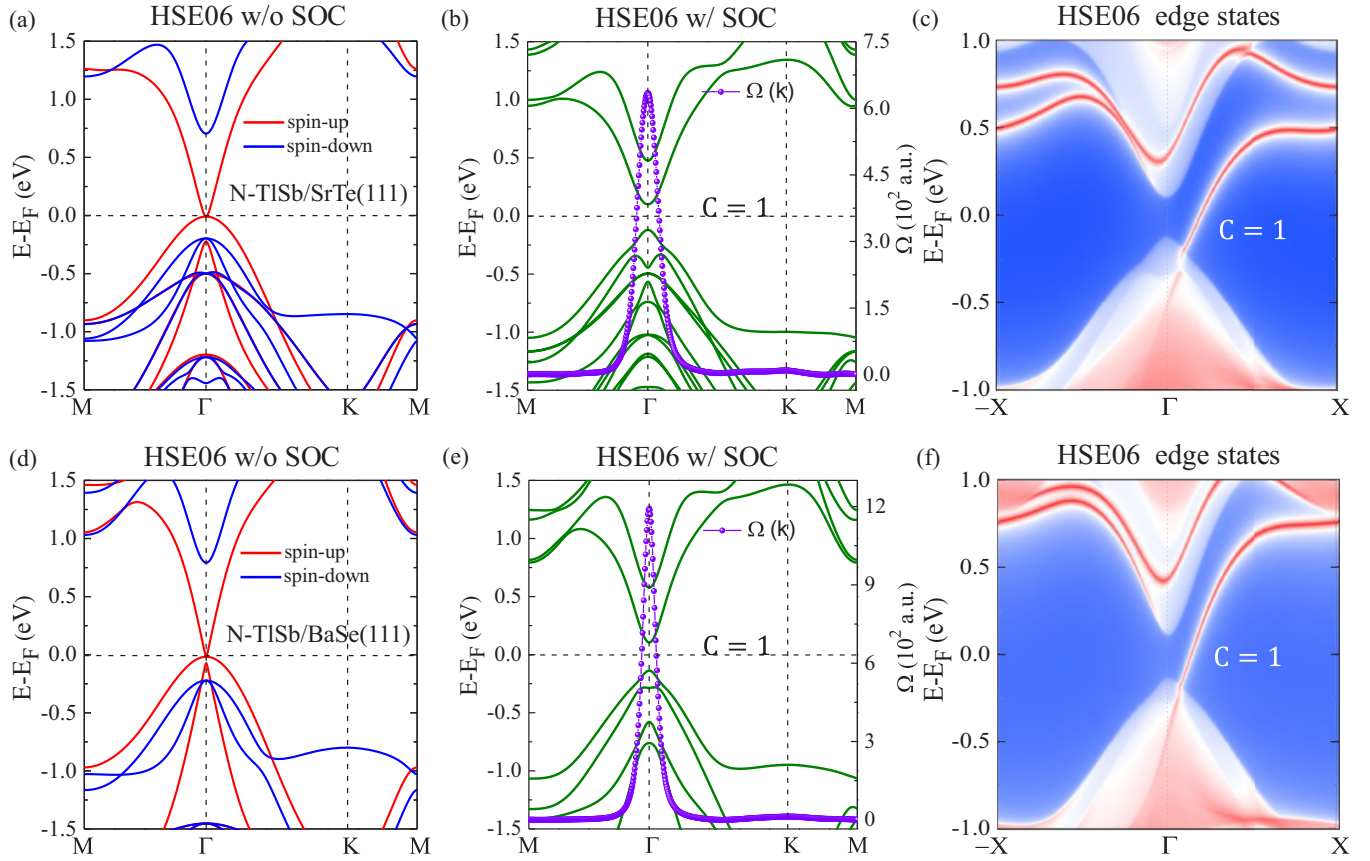


FIG. 5. (a) Spin-polarized band structure of N-TiSb/SrTe(111) with FM state obtained by using HSE06 calculations. The red and blue curves denote the spin-up and spin-down bands, respectively. (b) The corresponding band structure of (a) with the consideration of SOC. (c) The calculated edge states of semi-infinite N-TiSb/SrTe(111). (d)–(f) The same as (a)–(c), except that the calculated material system is the N-TiSb/BaSe(111) with FM state. The violet dots in (b) and (e) denote the calculated Berry curvatures.

both N-TiSb/SrTe(111) and N-TiSb/BaSe(111) are spin-gapless half-metallic states, similar to N-TiSb/BaTe(111) (Fig. 2(ii)). When SOC is considered, the global band gaps of 220.0 meV and 243.6 meV are opened around the non-Dirac Γ point for N-TiSb/SrTe(111) and N-TiSb/BaSe(111), respectively, as shown in Figs. 5(b) and 5(e). The calculated Berry curvatures $\Omega(\mathbf{k})$, Chern number ($C = 1$), and topologically nontrivial edge states shown in Figs. 5(b), 5(c) and 5(e), 5(f) demonstrate that the SOC-induced band gaps in N-TiSb/SrTe(111) [Fig. 5(b)] and N-TiSb/BaSe(111) [Fig. 5(e)] are both the non-Dirac QAH states. Due to the large topologically nontrivial gaps in N-TiSb/SrTe(111) and N-TiSb/BaSe(111), the obtained non-Dirac QAH states will not be destroyed by a small external magnetic field. The SOC-induced non-Dirac QAH gaps in Type-A N-TiSb/SrTe(111) and N-TiSb/BaSe(111) with GGA-PBE calculations are 51.8 meV and 151.2 meV, respectively, as shown in Fig. S6 in the SM [68]. Furthermore, Figs. S7 and S8 in the SM [68] demonstrate that the large-gap non-Dirac QAH states can also be achieved in the FM states of N-TiSb/SrTe(111) and N-TiSb/BaSe(111) with Type-B configuration.

We further investigate the electronic structures and topological properties of the TiSb/BaTe(111), TiSb/SrTe(111), and TiSb/BaSe(111) material systems functionalized with hydrogen or halogens. In contrast to nitrogen functionalization, our first-principles calculations reveal that neither

hydrogen nor halogen functionalization induces magnetism in TiSb/BaTe(111), TiSb/SrTe(111), and TiSb/BaSe(111). The calculated band structures and Z_2 topological invariants, shown in Figs. S9–S13 in the SM [68], demonstrate that the X-TiSb/BaTe(111), X-TiSb/SrTe(111), and X-TiSb/BaSe(111) ($X = \text{H, F, Cl, Br, I}$) systems are large-gap QSH insulators.

IV. CONCLUSIONS

In summary, the electronic structures and topological properties of the heterostructure of TiSb monolayer epitaxially grown on the nonmagnetic BaTe(111) surface are systematically investigated by using first-principles calculations. The degenerate energy point with quadratic non-Dirac band dispersion appears at the Γ point in TiSb/BaTe(111) without including SOC. The two non-Dirac bands around the E_F are mainly contributed by Sb and Tl p_x/p_y orbitals. After including SOC, a large-gap (331.0 meV) QSH state is realized around the non-Dirac Γ point in TiSb/BaTe(111). However, TiSb/BaTe(111) is in a metallic state due to the E_F is not located inside the QSH gap. Remarkably, the magnetism can be introduced in the TiSb/BaTe(111) heterostructure through nitrogen functionalization. The magnetic ground state of N-TiSb/BaTe(111) is the FM state, and its stability can be enhanced by applying a small vertical external magnetic field. Before considering SOC, N-TiSb/BaTe(111) is in a

spin-gapless half-metallic state, with the two spin-up non-Dirac bands being precisely degenerate at the E_F and the Γ point. After SOC is considered, a large global band gap of 258.0 meV is opened around the spin-polarized non-Dirac degenerate Γ point, indicating the realization of an intrinsic large-gap non-Dirac QAH state with significant Rashba SOC in the N-TiSb/BaTe(111) system. The band evolution and topological mechanism of the spin-polarized non-Dirac bands with atomic and Rashba SOC are analyzed using a $k \cdot p$ model. Additionally, our calculations show that the large-gap non-Dirac QAH states can also be achieved in the FM states of the N-TiSb/SrTe(111) and N-TiSb/BaSe(111) systems. Our findings demonstrate that nitrogen functionalized TiSb monolayers epitaxially grown on nonmagnetic insulating substrates are promising for the experimental exploration of the

large-gap QAH effect and hold potential for future QAH device applications.

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J.C.S. and J.X.L. contributed equally to this work.

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